ANDELA GEN AI ENGINEERING BOOTCAMP CAPSTONE PROJECT

Knowledge-base Search Solution

Advanced Retrieval Augmented Generation

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Project Vision

The Challenge

Existing legal AI research tools marketed as “hallucination-free” still generate inaccurate or fabricated legal information in 17%–33% of cases, creating an urgent need for a more reliable Advanced RAG system for accurate, citation-grounded legal research

Magesh et al., (2025)

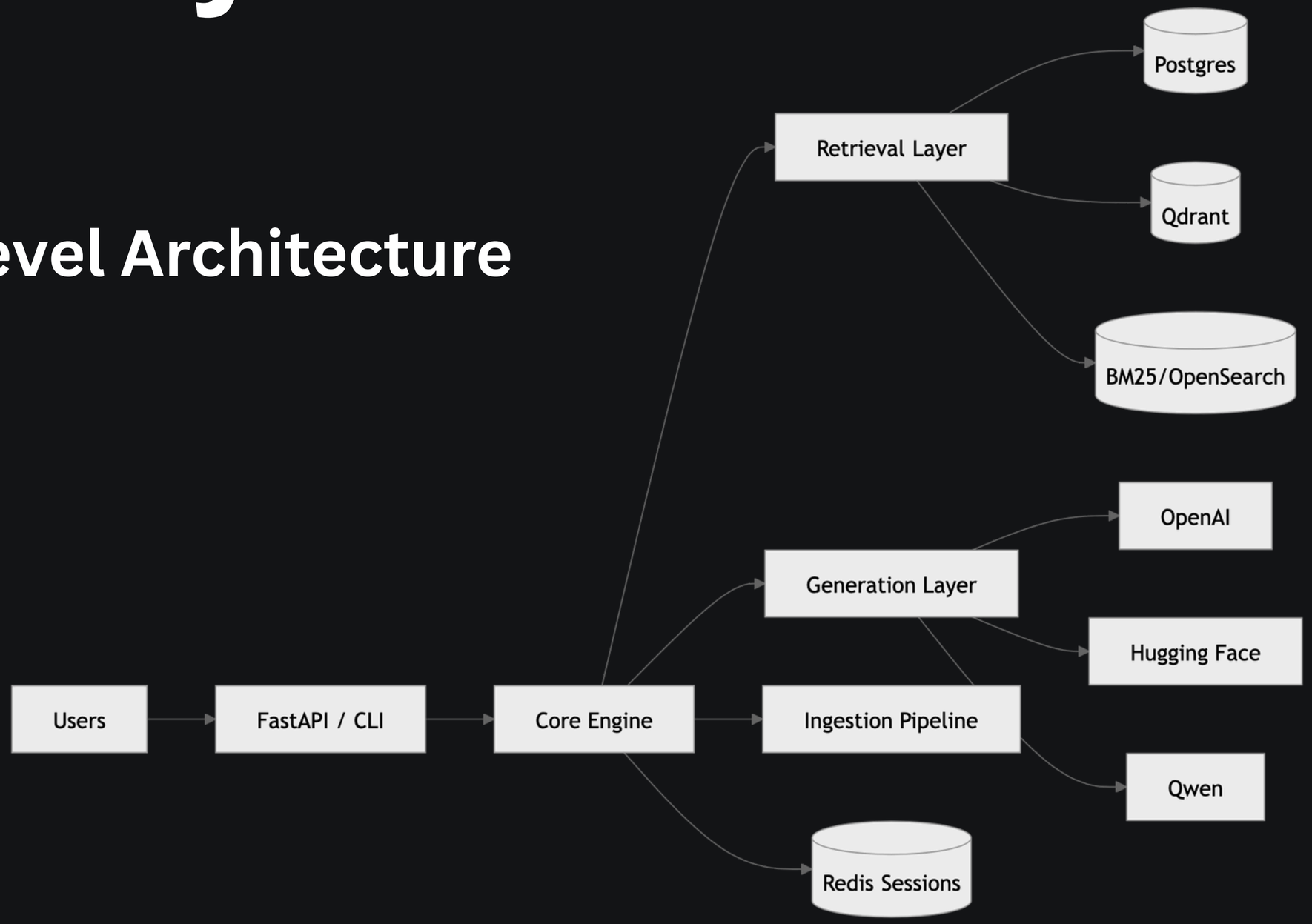
The Solution

Knowledge-base RAG solution with:

- Hybrid Retrieval & Re-ranking
- Access Control & Sensitivity Lane Enforcement
- Parent–Child Context Expansion
- Extractive Prompting with Mandatory Citations
- Faithfulness Verification & Confidence Scoring
- Refusal, Guardrails & Evaluation Calibration

System Architecture

High-Level Architecture



Technology Stack

Runtime and packaging

- Language: Python 3.10+
- App entry: click CLI (kb command), FastAPI + uvicorn[standard] for the HTTP server.

Core Libraries

- Config and validation: Pydantic v2, pydantic-settings
- HTTP: requests, httpx (external APIs, Qdrant HTTP, HF, OpenAI).
- LLM / APIs: OpenAI Python SDK; Qwen, Hugging Face inference

Data & Retrieval

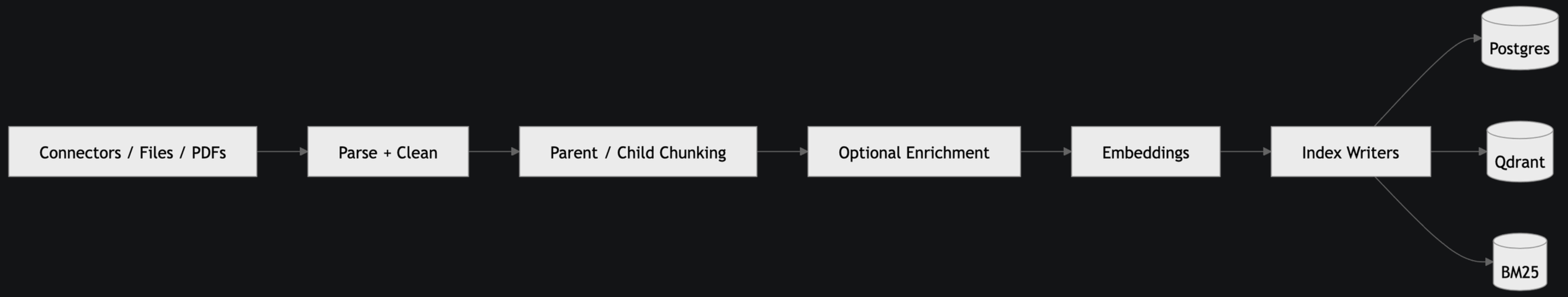
- Relational / document store: PostgreSQL
- Vector search: Qdrant
- Caching / sessions: Redis
- Lexical (sparse) retrieval: rank-bm25

Document & Text Processing

- PDFs: pypdf and pymupdf4llm (brings PyMuPDF for layout/PDF and preprocessing such as margin stripping).
- Structured / metadata: PyYAML.
- Token counting (chunking): tiktoken.

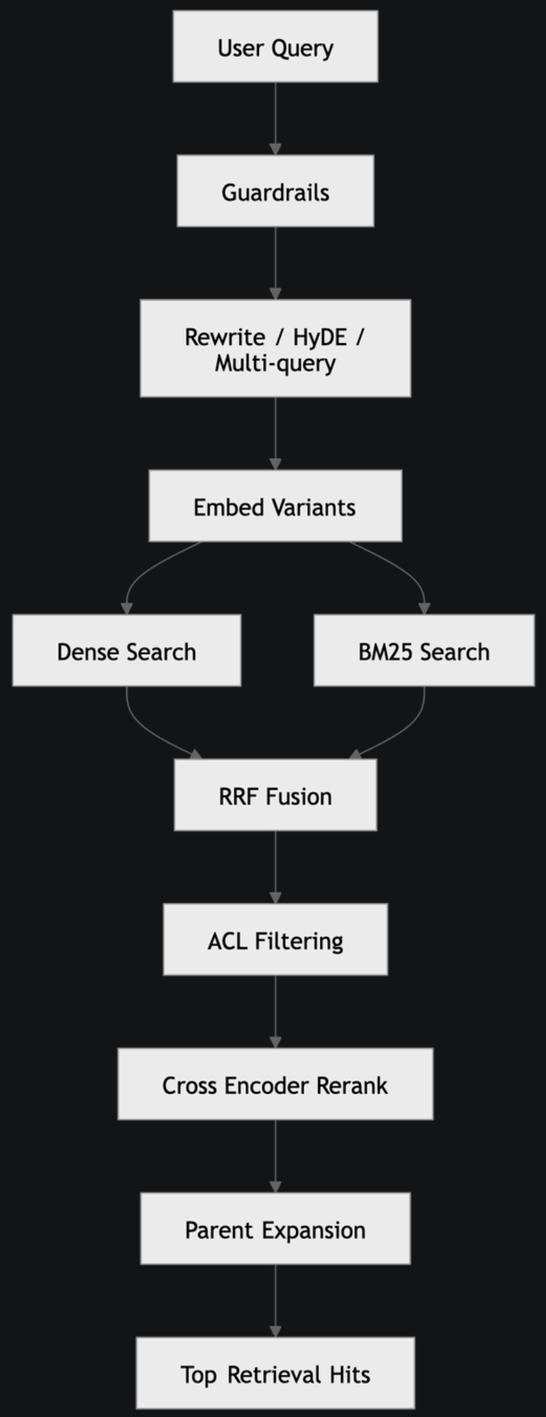
Ingestion Pipeline

Reliable, Idempotent Document Indexing



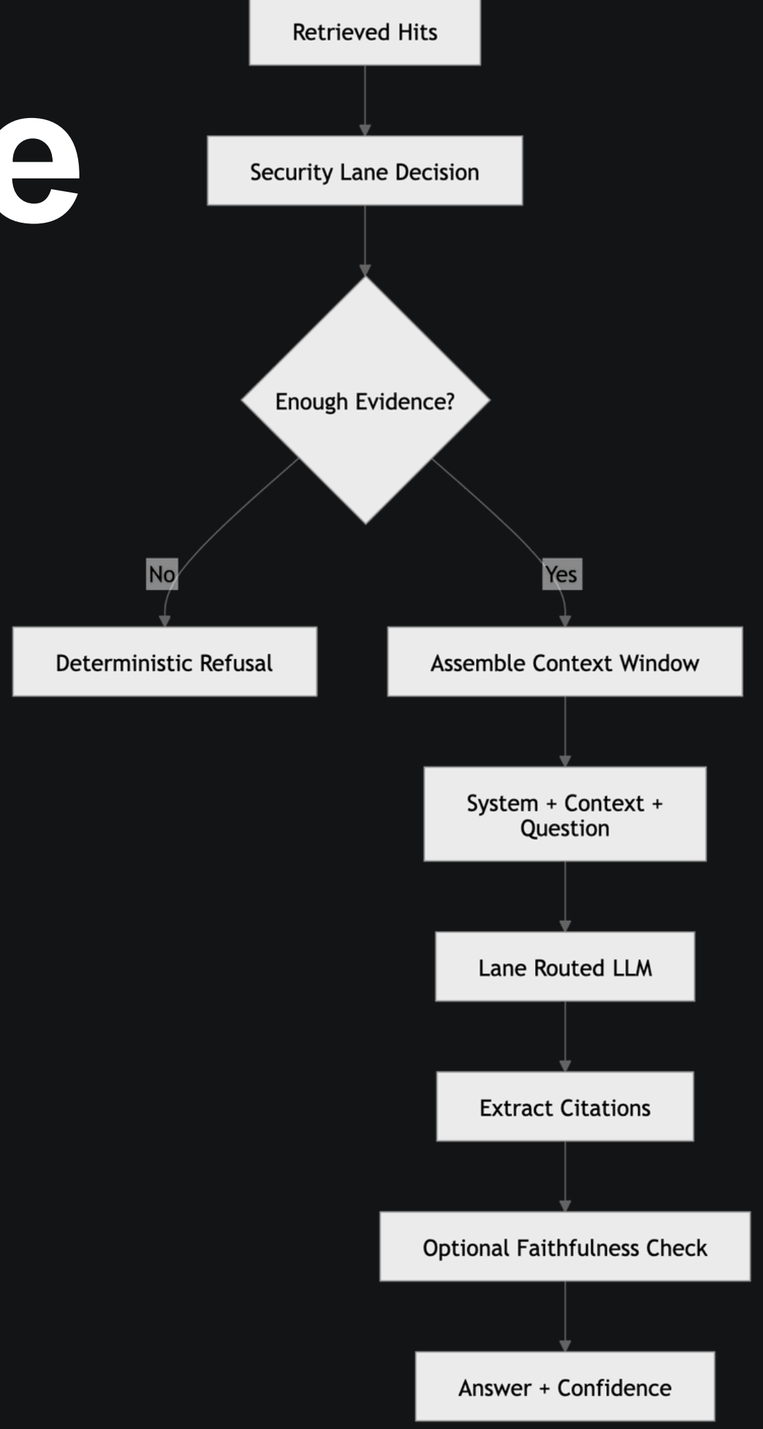
Retrieval Engine

Multi-Stage Hybrid Search Pipeline



Generation Pipeline

Controlled Answer Generation



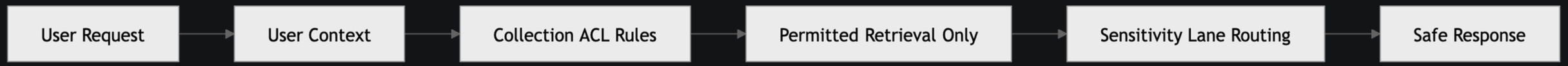
Hallucination Defence

How We Keep Answers Trustworthy

Layer	Control
Retrieval	Hybrid + rerank
Prompting	Cite sources only
Generation	Low temperature
Citation Parsing	Detect fake citations
NLI Verification	Test claims vs source
Confidence Score	Combined retrieval + support

Security + Enterprise Control

Built for Real Organizations



Included Controls:

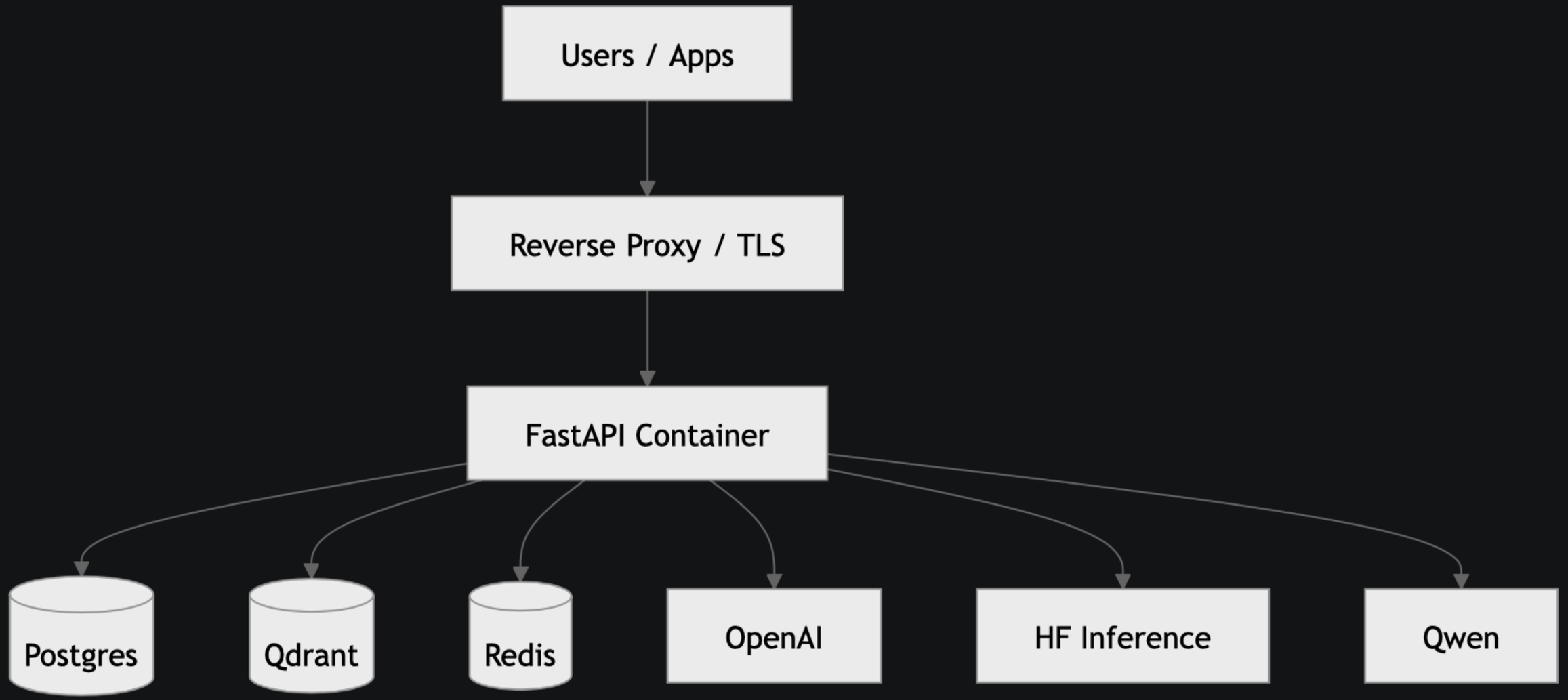
- Role-based collection access
- Query abuse guardrails
- Prompt injection heuristics
- Session memory with Redis

Interfaces + Operations

How Teams Use It

Interface	Purpose
CLI	Bulk ingest, search, eval
REST API	Integrate into apps
SSE Streaming	Real-time responses
Docker Compose	Fast local deployment

Deployment


Users / Apps
Reverse Proxy / TLS
FastAPI Container
Postgres
Qdrant
Redis
OpenAI
HF Inference
Qwen

Why Advanced RAG?

Basic RAG	This System
Single vector search	Hybrid retrieval
Blind generation	Citation-first
No confidence	Scored trust
No ACL	Enterprise-safe
One provider	Multi-provider lanes
Static pipeline	Extensible modules

Sample Query

Acme Knowledge Base
Hybrid search & cited answers

Anonymous (public on

USER

Act as

Anonymous (public only)

ACLs follow staff_directory (demo)

SESSION

Conversation ID

optional — for follow-up question

New session

Retrieval & generation

Context chunks (top-K)

8

☑ Rerank (cross-encoder)

Rewrite

Off

☐ Step-back variant

Ask | Search

YOU

How do I create an Acme workspace, and which regions are mentioned?

ANSWER

To create an Acme workspace, follow these steps:

1. Sign in to https://app.acme.com and click **New workspace**.
2. Choose a region close to your users — either `eu-west-1` or `us-east-1`.
3. Name the workspace and click **Create**.

Your workspace will be ready in about 30 seconds, and you will receive an API key by email to store in `~/.acme/credentials` for CLI access [6][7].

Citations

- [6] getting-started (src_sample_public)
- [7] getting-started (src_sample_public)

confidence 0.71 · gen 3929 ms · NLI 100% supported

Same Query, Different ACL

Acme Knowledge Base
Hybrid search & cited answers

David Lindqvist — dav

USER

Act as

David Lindqvist — david.lindqvis

ACLs follow staff_directory (demo)

SESSION

Conversation ID

optional — for follow-up question

New session

Retrieval & generation

Ask | Search

YOU

What must the hiring manager do at least 5 business days before a new employee starts?

ANSWER

The provided sources do not contain enough information to answer that. Information regarding hiring procedures, including actions required by a hiring manager before a new employee starts, is needed.

confidence 0.54 · gen 2025 ms · NLI 0% supported

YOU

What must the hiring manager do at least 5 business days before a new employee starts?

ANSWER

The hiring manager must file a new-hire ticket in the HR portal at least 5 business days in advance before the new employee starts [8].

Citations

- [8] onboarding-policy (src_sample_internal)

confidence 0.77 · gen 2902 ms · NLI 100% supported

Ask a question in natural language...

Monitoring

LangSmith

Personal / Tracing / acme-kb-search

acme-kb-search ID

Runs | Threads | Evaluators | Automations | Insights | Retention 14d | Dashboard | + New

Default View | Last 7 days | Traces Runs | Input Search input content | Reset | Save View

Add filter

Name	Input	Output	Error	Start Time	Latency	Dataset
smoke_test	smoke	true		4/25/2026, 5:...	0.23s	
smoke_test	smoke	true		4/24/2026, 6...	0.20s	
smoke_test	smoke	true		4/24/2026, 11...	0.19s	
smoke_test	smoke	true		4/24/2026, 1...	0.83s	
smoke_test	smoke	true		4/24/2026, 1...	1.87s	
smoke_test	smoke	true		4/24/2026, 1...	1.08s	
smoke_test	smoke	true		4/24/2026, 1...	1.56s	
smoke_test	smoke	true		4/24/2026, 1...	18.86s	

References

1. Magesh et al., 2025 – Hallucination-Free? Assessing the Reliability of Leading AI Legal Research Tools. Journal of Empirical Legal Studies. 22. 10.1111/jels.12413.

2. Gao et al., 2022 — HyDE (Precise Zero-Shot Dense Retrieval without Relevance Labels)

3. Zheng et al., 2023 — "Take a Step Back" (step-back prompting)